ic_autolab round 25 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T07:57:50Z at git `59a8d6377c9ab57e4f38fd3b60593d89962b8fd8`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-f461bc` (`python3 tools/pcc/pcc.py standing PCC-20261104-f461bc`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
20 **improvement**	wsum2-b15-n2	0.895 [0.862, 0.930]	0.899 [0.861, 0.938]	0.920 [0.856, 0.989]	sel < inc	4.860 / 5.083 / 7.092	5.428 / 5.656 / 7.709	1.502 / 1.609 / 2.571	3.2x / 3.2x / 2.8x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

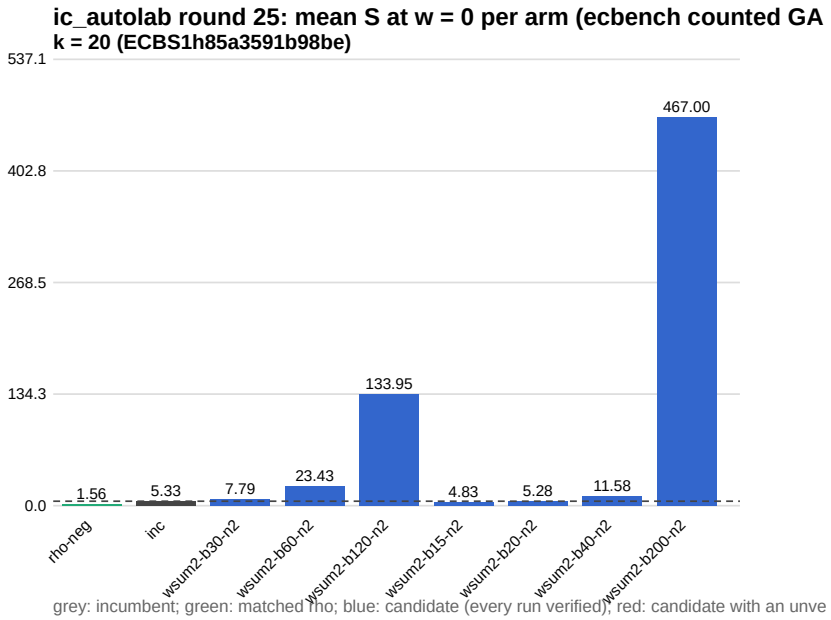
- k = 20: compare ops.status = ok, admissible = true; sel / inc = 0.8954 [0.862, 0.930] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

k = 20: session `ECBS1h85a3591b98be` (run exit 0, verify exit 0, receipt sha256 `fbe1dc22f8b2471f801f795a67a3d8b9ccb8aa884215b2e127e862547f8239b1`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	- -	`rho.negation`	32	32	1.558 / 1.670 / 2.682	matched reference
`inc`	wsum-b5	3 `prime-projected-abscissa:size=3,scan_cap=48`	`wsum:coef=5,free=1`	32	32	5.332 / 5.549 / 7.496	incumbent
`wsum2-b30-n2`	wsum2-b30	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=30,free=1`	32	32	7.786 / 7.854 / 8.471	cross-size transfer: wsum2-b30-n2 is the k = 24 incumbent at N / N_a = 0.025
`wsum2-b60-n2`	wsum2-b60	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=60,free=1`	32	32	23.431 / 23.573 / 24.847	cross-size transfer: wsum2-b60-n2 is the k = 28 incumbent at N / N_a = 0.010
`wsum2-b120-n2`	wsum2-b120	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=120,free=1`	32	32	133.949 / 141.785 / 212.308	cross-size transfer: wsum2-b120-n2 is the k = 32

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
							incumbent at N / N_a = 0.004
`wsum2-b15-n2`	wsum2-b15	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=15,free=1`	32	32	4.832 / 5.055 / 7.059	exploration: family wsum2-b15 untried at k = 20, N = 0.0001 x N_a
`wsum2-b20-n2`	wsum2-b20	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=20,free=1`	32	32	5.281 / 5.410 / 6.571	exploration: family wsum2-b20 untried at k = 20, N = 0.0001 x N_a
`wsum2-b40-n2`	wsum2-b40	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=40,free=1`	32	32	11.578 / 11.621 / 12.007	exploration: family wsum2-b40 untried at k = 20, N = 0.0001 x N_a
`wsum2-b200-n2`	wsum2-b200	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=200,free=1`	32	32	467.001 / 503.658 / 833.571	exploration: family wsum2-b200 untried at k = 20, N = 0.0001 x N_a



train screen

Selection

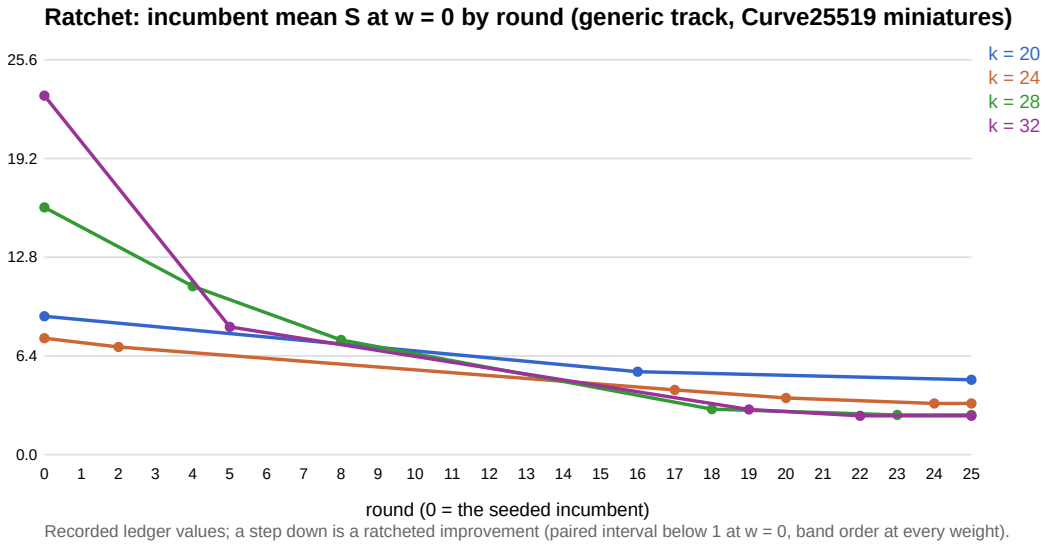
- k = 20: challenger `wsum2-b15-n2` (Pareto alternative at w = 1: `wsum2-b20-n2`)

Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
20	`ECBS1hf759e8a6c6f9`	0	0	`63cb936fe0d30b1d80410de1b50238acad52b8813067f5211f59d3057cc7132f`	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`wsum2-b15-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=15,free=1`)	25	4.860 / 5.083 / 7.092	3.2x / 3.2x / 2.8x
24	`wsum2-b30-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=30,free=1`)	24	3.318 / 3.573 / 5.863	2.7x / 2.7x / 2.5x
28	`wsum2-b60-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=60,free=1`)	23	2.576 / 2.793 / 4.746	2.8x / 2.7x / 2.6x
32	`wsum2-b120-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=120,free=1`)	22	2.519 / 2.759 / 4.915	2.7x / 2.7x / 2.7x



ratchet

Evidence and deviations

- ecbench binary sha256: `e7242bc192bda8c8da9e64895afe943e8aa47cfa37bd490d656e2181758e9533`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.